JEE 2009
Paper I

PART I : CHEMISTRY

SECTION – I

Single Correct Choice Type

This section contains 8 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

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1. Given that the abundances of isotopes ^{54}Fe, ^{56}Fe and ^{57}Fe are 5%, 90% and 5%, respectively, the atomic mass of Fe is

(A) 55.85 (B) 55.95 (C) 55.75 (D) 56.05

Answer ○ ● ○ ○

(A) (B) (C) (D)

2. The term that corrects for the attractive forces present in a real gas in the van der Waals equation is

(A) nb (B) $\frac{an^2}{V^2}$ (C) $-\frac{an^2}{V^2}$ (D) $-nb$

Answer ○ ● ○ ○

(A) (B) (C) (D)

3. Among the electrolytes Na_2SO_4, $CaCl_2$, $Al_2(SO_4)_3$ and NH_4Cl, the most effective coagulating agent for Sb_2S_3 sol is

(A) Na_2SO_4 (B) $CaCl_2$ (C) $Al_2(SO_4)_3$ (D) NH_4Cl

Answer ○ ○ ● ○

(A) (B) (C) (D)

4. The Henry's law constant for the solubility of N_2 gas in water at 298 K is 1.0×10^5 atm. The mole fraction of N_2 in air is 0.8. The number of moles of N_2 from air dissolved in 10 moles of water at 298 K and 5 atm pressure is

(A) 4.0×10^{-4} (B) 4.0×10^{-5} (C) 5.0×10^{-4} (D) 4.0×10^{-6}

Answer ● ○ ○ ○

(A) (B) (C) (D)

5. The reaction of P_4 with **X** leads selectively to P_4O_6. The **X** is

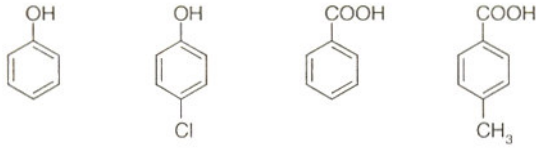
(A) Dry O_2 (B) A mixture of O_2 and N_2

(C) Moist O_2 (D) O_2 in the presence of aqueous NaOH

Answer ○ ● ○ ○

(A) (B) (C) (D)

6. The correct acidity order of the following is



(I) (II) (III) (IV)

(A) (III) > (IV) > (II) > (I)
(B) (IV) > (III) > (I) > (II)
(C) (III) > (II) > (I) > (IV)
(D) (II) > (III) > (IV) > (I)

Answer ● ○ ○ ○

(A) (B) (C) (D)

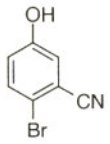
7. Among cellulose, poly(vinyl chloride), nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest is

(A) Nylon
(B) Poly(vinyl chloride)
(C) Cellulose
(D) Natural Rubber

Answer ○ ○ ○ ●

(A) (B) (C) (D)

8. The IUPAC name of the following compound is



(A) 4-Bromo-3-cyanophenol
(B) 2-Bromo-5-hydroxybenzonitrile
(C) 2-Cyano-4-hydroxybromobenzene
(D) 6-Bromo-3-hydroxybenzonitrile

Answer ○ ● ○ ○

(A) (B) (C) (D)

SECTION – II

Multiple Correct Choice Type

This section contains 4 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

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9. The correct statement(s) regarding defects in solids is(are)

(A) Frenkel defect is usually favoured by a very small difference in the sizes of cation and anion
(B) Frenkel defect is a dislocation defect
(C) Trapping of an electron in the lattice leads to the formation of F-center
(D) Schottky defects have no effect on the physical properties of solids

Answer ○ ● ● ○ **OR** ○ ○ ● ○

(A) (B) (C) (D) (A) (B) (C) (D)

10. The compound(s) that exhibit(s) geometrical isomerism is(are)

(A) $[Pt(en)Cl_2]$ (B) $[Pt(en)_2]Cl_2$

(C) $[Pt(en)_2Cl_2]Cl_2$ (D) $[Pt(NH_3)_2Cl_2]$

Answer ○ ○ ● ●

(A) (B) (C) (D)

11. The compound(s) formed upon combustion of sodium metal in excess air is(are)

(A) Na_2O_2 (B) Na_2O (C) NaO_2 (D) NaOH

Answer ● ○ ○ ○ **OR** ● ● ○ ○

(A) (B) (C) (D) (A) (B) (C) (D)

12. The correct statement(s) about the compound $H_3C(HO)HC-CH=CH-CH(OH)CH_3$ (**X**) is(are)

(A) The total number of stereoisomers possible for **X** is 6

(B) The total number of diastereomers possible for **X** is 3

(C) If the stereochemistry about the double bond in **X** is *trans*, the number of enantiomers possible for **X** is 4

(D) If the stereochemistry about the double bond in **X** is *cis*, the number of enantiomers possible for **X** is 2

Answer ● ○ ○ ●

(A) (B) (C) (D)

SECTION – III

Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 13 to 15

p-Amino-*N*,*N*-dimethylaniline is added to a strongly acidic solution of **X**. The resulting solution is treated with a few drops of aqueous solution of **Y** to yield blue coloration due to the formation of methylene blue. Treatment of the aqueous solution of **Y** with the reagent potassium hexacyanoferrate(II) leads to the formation of an intense blue precipitate. The precipitate dissolves on excess addition of the reagent. Similarly, treatment of the solution of **Y** with the solution of potassium hexacyanoferrate(III) leads to a brown coloration due to the formation of **Z**.

13. The compound **X** is

(A) $NaNO_3$ (B) NaCl (C) Na_2SO_4 (D) Na_2S

Answer ○ ○ ○ ●

(A) (B) (C) (D)

14. The compound **Y** is

(A) $MgCl_2$ (B) $FeCl_2$ (C) $FeCl_3$ (D) $ZnCl_2$

Answer ○ ○ ● ○

(A) (B) (C) (D)

15. The compound **Z** is

(A) $Mg_2[Fe(CN)_6]$ (B) $Fe[Fe(CN)_6]$

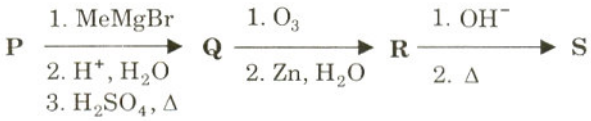
(C) $Fe_4[Fe(CN)_6]_3$ (D) $K_2Zn_3[Fe(CN)_6]_2$

Answer ○ ● ○ ○

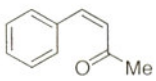
(A) (B) (C) (D)

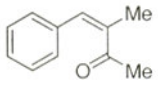
Paragraph for Question Nos. 16 to 18

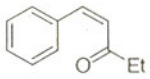
A carbonyl compound **P**, which gives positive iodoform test, undergoes reaction with MeMgBr followed by dehydration to give an olefin **Q**. Ozonolysis of **Q** leads to a dicarbonyl compound **R**, which undergoes intramolecular aldol reaction to give predominantly **S**.

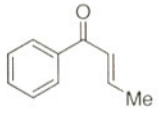


16. The structure of the carbonyl compound **P** is

(A) 

(B) 

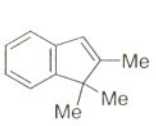
(C) 

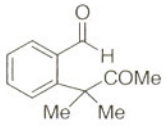
(D) 

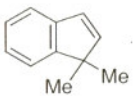
Answer ○ ● ○ ○

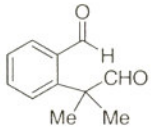
(A) (B) (C) (D)

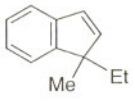
17. The structures of the products **Q** and **R**, respectively, are

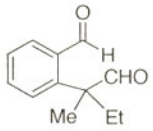
(A) 

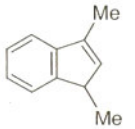
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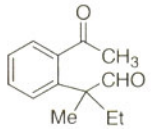
(B) 

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(C) 

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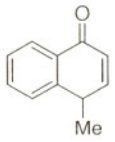
(D) 

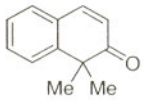
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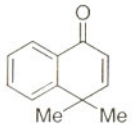
Answer ● ○ ○ ○

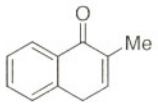
(A) (B) (C) (D)

18. The structure of the product **S** is

(A) 

(B) 

(C) 

(D) 

Answer

(A) (B) (C) (D)

SECTION – IV

Matrix – Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled p, q, r, s and t. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – p, s and t; B – q and r; C – p and q; and D – s and t; then the correct darkening of bubbles will look like the following.

	p	q	r	s	t
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

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19. Match each of the diatomic molecules in **Column I** with its property/properties in **Column II**.

Column I	**Column II**
(A) B_2	(p) Paramagnetic
(B) N_2	(q) Undergoes oxidation
(C) O_2^-	(r) Undergoes reduction
(D) O_2	(s) Bond order ≥ 2
	(t) Mixing of 's' and 'p' orbitals

Answer

	p	q	r	s	t
A	●	●	●	○	●
B	○	●	●	●	●
C	●	●	●	○	○
D	●	●	●	●	○

20. Match each of the compounds in **Column I** with its characteristic reaction(s) in **Column II**.

Column I	**Column II**
(A) $CH_3CH_2CH_2CN$	(p) Reduction with $Pd-C/H_2$
(B) $CH_3CH_2OCOCH_3$	(q) Reduction with $SnCl_2/HCl$
(C) $CH_3-CH=CH-CH_2OH$	(r) Development of foul smell on treatment with chloroform and alcoholic KOH
(D) $CH_3CH_2CH_2CH_2NH_2$	(s) Reduction with diisobutylaluminium hydride (DIBAL-H)
	(t) Alkaline hydrolysis

Answer

	p	q	r	s	t
A	●	●	○	●	●
B	○	○	○	●	●
C	●	○	○	○	○
D	○	○	●	○	○

PART II : MATHEMATICS

SECTION – I

Single Correct Choice Type

This section contains 8 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

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21. Let $P(3, 2, 6)$ be a point in space and Q be a point on the line

$$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(-3\hat{i} + \hat{j} + 5\hat{k}).$$

Then the value of μ for which the vector $\overrightarrow{PQ}$ is parallel to the plane $x - 4y + 3z = 1$ is

(A) $\frac{1}{4}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $-\frac{1}{8}$

Answer ● ○ ○ ○

(A) (B) (C) (D)

22. Tangents drawn from the point $P(1, 8)$ to the circle

$$x^2 + y^2 - 6x - 4y - 11 = 0$$

touch the circle at the points A and B. The equation of the circumcircle of the triangle PAB is

(A) $x^2 + y^2 + 4x - 6y + 19 = 0$ (B) $x^2 + y^2 - 4x - 10y + 19 = 0$

(C) $x^2 + y^2 - 2x + 6y - 29 = 0$ (D) $x^2 + y^2 - 6x - 4y + 19 = 0$

Answer ○ ● ○ ○

(A) (B) (C) (D)